

Learning to choose, then navigate

Experiment 2 asks whether an explicitly trained controller can associate a visual pattern with reward and learn routes to food in an editable maze. A two-dimensional fly avatar makes the decisions visible. This is an engineering experiment that uses a fixed connectome-derived sensory encoder; it is not a claim that biological fly memories have been reconstructed.

Prospective status

Written on 20 September 2026 before implementation of this learning mechanism and before its training outcomes were observed. Numerical verification and exploratory development follow. This is a dated local protocol, not an external preregistration or a guarantee of success.

Preserve Experiment 1

Experiment 1 keeps its current launcher, UI, dynamics, graph and PDFs. The local Git tag exp-1-v0.2 points to commit 53dff10. Experiment 2 has setup_exp_2.sh, its own service and port 8767, separate run records and an independent interface. The existing Mac PLAN folder gains only this exp_2.pdf.

The controller learns. Connectome edges and neural model parameters remain fixed. A successful food-finding route alone does not demonstrate biological learning, cognitive understanding or an advantage from the fly wiring.

The first maze task

The arena is a 9 by 9 grid. Walls block movement. The user can edit walls, the starting cell and two food sites. The agent has four actions: north, east, south and west. One action moves one grid cell. The rendered body is a schematic avatar, not a biomechanical fly.

The two sites display Pattern A (vertical bars) and Pattern B (horizontal bars). Their assignment swaps across episodes using a reproducible, balanced schedule. The rewarded pattern is configured before the experiment; the controller is not given the reward label. Contact with either food site ends the episode.

A correct food contact yields +1; the other pattern yields -0.25. Each ordinary movement costs -0.01. An attempted manual move into a wall costs -0.04. Autonomous policies see adjacent walls and choose among legal moves. The episode times out after 120 actions.

What the agent observes

Its engineered grid position, legal adjacent movements, and both food-site pattern signals are available throughout an episode. The visual beacons are readable across the arena and are not occluded by walls. The controller receives decoded sensory tokens; it receives no shortest path, food payoff label or distance-to-goal oracle.

This first task is route learning with visible beacons and perfect grid localization. It does not yet model an unknown maze, realistic fly vision, smell, fog of war or a general search strategy. Transfer tests explicitly check the limits of route memory.

From patterns to neural signals

In real-data mode, load the same 166,700-node MaleCNS-derived graph and fixed LIF equations used in Experiment 1. Feed a 32 by 24 pattern into mapped L1 columns on both sides, using the existing exploratory coordinate mapping and a drive of 0.6 mV per tick. Each pattern probe starts from reset and runs for 40 ms.

Pool measured L1 spike rates into a 4 by 4 grid for each eye: 32 neural features. Preserve the feature vectors, whole-network counts, voltage summaries, mapping identity, model parameters and graph checksums. Demonstrate distinguishable, finite responses before allowing neural-cue training.

The two known pattern responses become fixed calibration templates. Classification chooses the nearest response template. This is an engineered decoder calibrated on the two stimuli; it is not a discovered natural motor or recognition circuit.

Exact cached probes, not a continuous brain

Because this task repeats only two deterministic images, compute each reset response once and reuse its recorded features by image identity. The UI labels the neural plot as a cached probe. No new full-brain simulation occurs on every movement, and no continuous neural state connects maze steps.

On a Mac the portable default uses the explicitly synthetic teaching graph and CPU solver. On gx10-b the actual MaleCNS probes run on CUDA. The small Q-learning controller and maze run on the same server CPU. Never display cached traces as current whole-brain activity.

The learning rule

Use tabular Q-learning. $Q(s,a)$ estimates the discounted future return of action a in state s . Here s contains the grid position and the decoded pattern at each site. The Q table has four sensory combinations, 81 positions and four actions. Walls and the reward rule are fixed within a training experiment.

One-step update

$Q(s,a)$ becomes $Q(s,a) + \alpha * [r + \gamma * \max Q(\text{next_s}, \text{legal_action}) - Q(s,a)]$. For a terminal food contact or timeout, use r alone as the target. The learned quantities are entries in this external table; no synapse in the connectome is updated. [1]

Defaults: $\alpha = 0.25$; $\gamma = 0.95$. Exploration probability ϵ is $\max(0.05, 0.8 * 0.99^{\text{episode}})$. During training, choose a uniformly random legal action with probability ϵ ; otherwise choose a maximum- Q legal action, breaking ties reproducibly at random. Evaluation uses ϵ zero and never updates Q .

The step cost encourages shorter successful routes without revealing their direction. Exploration can initially look inefficient. A finite run with a fixed learning rate does not inherit an unconditional mathematical guarantee of optimality.

Resetting a trial preserves Q . Starting a new experiment or applying edited geometry starts fresh Q tables and a new run record. This prevents mixing incompatible reward tasks in one learning curve.

Controls that matter

The interface shows an editable maze, a six-legged fly avatar, its trajectory, visited cells and food patterns. The side panels show current reward, episode, exploration, action values, neural template features and the learning curve.

Train, Pause and Step

Train adds a chosen number of episodes to a bounded training queue. Speed controls display pacing in actions per second, not a biological time constant or a learning parameter. Pause keeps Q and the current trial. Step makes one learner decision and applies its update.

Manual movement

The arrow controls move the avatar while paused. Manual trials do not update the trained Q table or enter the autonomous training-success metric. Reset trial starts a clean autonomous trial while retaining learned Q. Finish/save preserves the experiment checkpoint.

Edit the arena

Select Wall, Erase, Start, Food site 1 or Food site 2, then click cells. Draft edits take effect only when Apply new experiment is pressed. Validation requires a closed boundary, distinct open markers and routes from the start to both sites without needing to pass through the other terminal site.

New experiment and reward reversal

New maze drafts a different layout. The rewarded-pattern selector and seed apply when creating a new experiment. Changing the reward identity resets learning; studying reversal while retaining memory requires a separate future protocol amendment.

Controlled evaluation

Every training episode has a matching image-cue control episode. That controller sees the same grid position and legal movements and uses the same Q-learning rule, initial values, exploration schedule, reward scheme and episode random seed. It recognizes the two rendered patterns directly, bypassing the neural simulation.

The image-cue control receives equivalent pattern information. If its decoded tokens match the neural templates, equal performance is expected. That outcome means the maze task and Q-learning worked, without evidence that the connectome contributed an advantage.

Evaluate freezes a policy checkpoint and runs all three agents: trained neural-cue Q, trained image-cue Q, and uniformly random legal movement. Use a disjoint seed namespace for evaluation. Report the trained maze separately from newly generated mazes with the same grid size and marker conventions.

Default evaluation uses 30 episodes per environment group and per agent. Repeat food-pattern swaps in balanced pairs. No weight changes occur during either familiar-maze or unseen-maze evaluation. Evaluation results cannot be appended to the training curve as if they were training trials.

Unseen mazes may expose strong overfitting because Q is indexed by coordinates. Failure to transfer is a valid result. This first protocol cannot establish a universal or best maze-scanning strategy.

What to measure

Primary outcomes: fraction of episodes reaching the rewarded pattern and mean return across all episodes. Also report steps, food mistakes and timeouts. Show a rolling training success rate using the last 20 completed autonomous episodes; label its actual sample count.

During evaluation, preserve per-trial outcomes and report successes divided by n with a Wilson 95 percent interval. This interval describes the finite trial sample, not variation across training seeds or hardware. A single training seed is an exploratory demonstration.

For successful episodes, shortest-path efficiency is shortest allowed start-to-goal length divided by steps taken. The shortest-path calculation belongs only to the evaluator; the policy never receives it. Paths through the wrong terminal food cell are excluded.

For a stronger follow-up, use at least five independently trained seeds, summarize per-seed results, freeze hyperparameters before evaluating and test on additional held-out layouts. Do not claim significance from hundreds of steps of the same policy as if each were an independent training experiment.

Development acceptance

The system must demonstrate actual reward-driven Q updates, checkpoint restoration, unchanged Q during evaluation, correctly excluded manual trials and preserved Experiment 1. Record favorable and unfavorable results. Performance comparisons must name the dataset, trained component and control.

Installation and two-machine use

From a cloned project, run `sh setup_exp_2.sh`. This creates or reuses the isolated runtime, prepares the labeled synthetic graph and starts Experiment 2 on `http://127.0.0.1:8767/`. Use `--no-start` to install and verify without launching. It does not start or modify Experiment 1.

For real data on a CPU, use `sh setup_exp_2.sh --dataset male-cns --backend cpu`. For a compatible Linux NVIDIA system, use `--dataset male-cns --backend cuda`. Python 3.11-3.13 and the same pinned dependencies apply. The full-data memory and download requirements remain those in the handbook.

In this lab, keep Experiment 1 on `gx10-a` at port 8765. Run Experiment 2 on `gx10-b` at port 8767 and forward only loopback to the Mac. The machines run separate experiments; no distributed simulation is claimed. Both receive matching source releases.

The launcher accepts `--port`, `--venv`, `--data-dir` and `--runs-dir`. Its default records directory is `runs/exp_2`. A busy port produces an actionable error; existing processes are not terminated. Cached neural probes are generated from the selected graph, never substituted from another dataset.

The user-facing maze is rendered in the Mac browser. Numerical probes and all experiment state are owned by the selected server. Refreshing the browser does not reset learning. Two tabs of one service share the same experiment.

Records, checkpoints and restoration

Each experiment gets a unique directory containing `source_manifest.json`, `manifest.json`, `config.json`, `commands.jsonl`, `episodes.jsonl`, `evaluations.json` and `checkpoint.npz`. Neural probe vectors and summaries are copied into the record with graph and mapping hashes; source code is preserved once per source identity.

The checkpoint stores both learned Q tables, completed episode count, validated layout/configuration and encoder identity. It is saved after each completed episode and on pause/save. Loading verifies dataset/encoder identity and opens a new run linked to the parent checkpoint, preserving the original record.

Checkpoint restoration resumes from a fresh episode at the boundary. It does not restore a partially completed maze trajectory or a continuously running brain, because those are not the state represented by this checkpoint. The saved episode count preserves the exploration schedule.

Each evaluation records its policy digest, seeds, layouts and trial outcomes. Check Q arrays before and after evaluation to reject accidental training leakage. Manual interventions are marked separately. Partial or failed runs retain their evidence and an explicit status.

Reproduction

Preserve source archives, graph data/checksums, neural probe artifacts, seed/configuration and policy checkpoints. A replay of the external controller can reuse verified probe features; reproducing the neuroscience probe additionally needs the graph and numerical backend. Dataset downloads and run records remain outside Git.

Interpretation and next stages

A higher reward rate on the training maze shows a controller adapted to this finite task. It does not show that the MaleCNS organism learned, that fly synapses changed, or that the avatar has natural fly behavior. The interface must keep the trainable controller and fixed neural encoder visibly separate.

A similar result for the image-cue baseline is informative. It identifies the ordinary learner as sufficient for the task. A claimed benefit from fly anatomy would need harder matched tests and appropriate wiring perturbations, with encoder information and training budgets controlled.

After this version: (1) test unseen-layout transfer, (2) add partial visibility and explicit agent memory, (3) consider an olfactory search task, or (4) introduce a stated mushroom-body-inspired plasticity rule. Each addition changes the scientific question and requires a dated amendment before evaluating its claims.

Published embodied fly platforms provide useful examples of sensory interfaces and trained controllers. They do not make our particular graph, dynamics or Q table biologically validated. Connectome-constrained visual work also fits physiological parameters; we are not adopting those fitted dynamics automatically. [2,3]

Success is measured improvement with reproducible records and appropriate controls. If the best observed strategy memorizes this maze, report memorization. If transfer or biological specificity is absent, report that explicitly.

References and protocol boundary

[01] Watkins and Dayan. Q-learning. Machine Learning (1992).

<https://www.gatsby.ucl.ac.uk/~dayan/papers/cjch.pdf>

[02] Wang-Chen et al. NeuroMechFly v2: simulating embodied sensorimotor control in adult Drosophila. Nature Methods (2024).

<https://www.nature.com/articles/s41592-024-02497-y>

[03] Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature (2024).

<https://www.nature.com/articles/s41586-024-07939-3>

[04] MaleCNS official download inventory.

<https://male-cns.janelia.org/download/>

The official MaleCNS release supplies anatomical data. The current project supplies assumed dynamics, direct L1 stimulation, calibrated template readout, grid localization, reward values and the external learning mechanism. These layers must remain distinguishable.

Protocol v0.1 is the pre-implementation specification. Implementation checks and observed development results belong in STATUS.md and run records. A later document revision must identify what was observed before it was written. Keep one current exp_2.pdf in PLAN and preserve prior source versions in Git.

Amendment: a fair comparison

21 September 2026. This amendment is written before implementing and observing the new trainable-circuit benchmark. The original 20 September maze runs were already observed: fixed-neural-cue and image-cue Q tables were identical, with 30/30 familiar successes and 8/30 on unseen layouts in one development run. That is evidence of no added value from the cached visual probes in that implementation.

Two different questions

Part A audits the existing Exp 2 mechanism across ten seeds and both rewarded patterns: does the measured neural encoder change any decoded observation, Q update or outcome compared with direct image cues? Both use identical episode assignments, random decisions, rewards, hyperparameters and training budgets.

Part B tests whether a reduced network constrained by actual fly connections can learn action values better than a Q table under identical collected experience. It does not train the complete 166,700-neuron LIF model. A separate benchmark preserves the existing UI, sessions and graph files.

Q-learning is a learning algorithm; a connectome is a connectivity structure. Our neural controllers also learn action values using temporal-difference learning. The comparison is tabular versus neural representation, and anatomical versus randomized connectivity - not biology versus all reinforcement learning.

Exactly what will be trained

Load the verified MaleCNS graph and its annotations. Identify Kenyon_Cell and MBON classes. Select 256 connected Kenyon cells using a fixed anatomy-only random seed, retain their recorded positive-weight connections to MBONs, and discard MBON rows without a selected connection. Save every body ID, edge mask and initial weight. No behavioral outcomes influence circuit selection.

This deliberately reduced feedforward rate model excludes recurrent, modulatory and other brain pathways. Recorded contact counts initialize positive strengths, normalized to unit incoming sum. The imported positive sign is a modeling assumption. It is not the unchanged full-brain spiking simulation and is not validated mushroom-body physiology.

All agents observe grid x/y, the same two pattern identities and four legal-move bits. No rewarded-pattern label, complete maze, goal label, distance-to-food or shortest path enters the observation. A fixed random projection encodes one-hot x/y and both pattern labels plus legal moves into 256 nonnegative Kenyon-like activities. This projection is engineered, not recorded fly sensory wiring.

MBON-like activity = $\tanh(W \text{ times KC activity})$. Four learned linear readouts produce north/east/south/west action values. In the plastic variant, reward-driven TD errors update existing entries of W and the readout; absent edges stay zero and strengths stay nonnegative. In the frozen-circuit variant, only the readout changes.

Updating modeled KC-to-MBON strengths is genuine parameter training in this reduced artificial network. It is not proof of biological synaptic plasticity. Backpropagation/Adam and the external action readout are engineered learning mechanisms.
[5]

Controls and equal experience

Controllers in Part B: (1) tabular Q-learning, (2) recorded circuit with a trainable readout and frozen internal weights, (3) recorded circuit with trainable internal weights and readout, (4) randomized circuit with the same training mechanism and number of parameters. A random-action policy is an evaluation reference.

The randomized control uses directed bipartite edge swaps preserving incoming and outgoing degrees. It retains the same selected cells, input projection, readout initialization, positive weight multiset per output cell and edge count. The source and randomized masks and their difference are recorded. This isolates topology more closely than an unmatched dense ANN.

For each of ten seeds and both rewarded patterns, collect exactly 20,000 environment transitions using uniform random legal actions on the default maze. Reset at terminal outcomes or the 120-step limit. This shared behavior dataset prevents one controller from receiving a more favorable learning history.

Each controller receives the same 2,000 replay batches of 64 transitions. Tabular Q uses alpha 0.25 and gamma 0.95. Neural Q uses Huber TD loss, Adam learning rate 0.001, gamma 0.95, a frozen target copy refreshed every 100 updates, gradient norm clipping at 5, and nonnegative internal weights clipped to 3. No test-result tuning is allowed within this protocol.

All neural controllers share the input projection and initial readout for a given seed. No pretraining or privileged labels are used. The full graph is immutable; learned reduced-circuit weights go into separate checkpoints. Equal experience and update counts do not imply equal wall time or parameter count; both are reported.

Evaluation and decision rules

Part A: ten training seeds, rewarded A and B, 400 training episodes per condition, 100 frozen trials on the familiar maze and 100 on unseen layouts. Compare decoded tokens and exact Q-table hashes, and save matched outcomes. The current encoder resets for each of its two 40 ms probes, and does not evolve during movement.

Part B: evaluate before training and after 500, 1,000 and 2,000 updates. At each checkpoint run 100 trials on the familiar maze and 100 on unseen layouts, using a separate RNG seed namespace. Pair assignments, mazes and action-tie seeds across agents. Reusing frozen test conditions supports learning curves, not hyperparameter selection.

Exploration and learning are off during evaluation. Preserve policy hashes before/after, and fail if they change. Use 50 fixed unseen layouts with both pattern assignments. Exact x/y and adjacent walls remain observable, so this tests limited layout transfer rather than natural fly navigation or full-map planning.

Primary endpoint: familiar-maze success after the fixed budget. Report unseen-layout success separately, along with mean return, steps, successful-route efficiency, training time and trainable parameter counts. Compute paired differences averaged over both rewarded patterns within each training seed, with a bootstrap 95% interval over the ten seed units.

Call a controller better only on the named metric and tested setting; an interval overlapping zero is inconclusive. An anatomical benefit requires improvement over the matched randomized circuit as well as relevant tabular results. A failed neural learner, a tie or poor transfer must remain visible. This small benchmark cannot establish superiority over all ML/RL models.

Reproduction and scope updates

The comparison runner will save a prospective config, source manifest/archive, verified graph identity, selected circuit IDs/masks/weights, raw shared transition data, replay seeds, checkpoints, learning curves, trial outcomes, timing and a machine-readable report. Records remain separate from the live maze experiment.

Full-graph visual probes run with the requested existing CPU/CUDA backend. The small reduced-circuit benchmark uses NumPy on the GX10 host CPU; it does not silently claim GPU training. Dependency pins and source archives identify the implementation. Development smoke runs are excluded from the reported benchmark.

The protocol and its fingerprint are saved before collecting new outcomes. Any deviations or later changes to training budgets must be dated and reported, not retroactively described as prospective. Original Exp 1 dynamics and the current interactive Exp 2 learning rule are preserved.

Current setup defaults have changed since the original pages were written: `setup.sh`, `setup_exp_2.sh` and `setup_explorer.sh` now select real MaleCNS data. Synthetic software fixtures require `--dataset demo` explicitly. This setup correction does not change the original experimental results.

[01] Watkins and Dayan. Q-learning. Machine Learning (1992).

<https://www.gatsby.ucl.ac.uk/~dayan/papers/cjch.pdf>

[03] Lappalainen et al. Connectome-constrained networks predict neural activity across the fly visual system. Nature (2024).

<https://www.nature.com/articles/s41586-024-07939-3>

[04] MaleCNS official download inventory.

<https://male-cns.janelia.org/download/>

[05] Learning with reinforcement prediction errors in a model of the Drosophila mushroom body (2021).

<https://www.nature.com/articles/s41467-021-22592-4>

No comparative result from the new plastic network exists at the time of this amendment. Observed outcomes will be added to STATUS.md and the separate run report after verification.

Observed results: 21 September

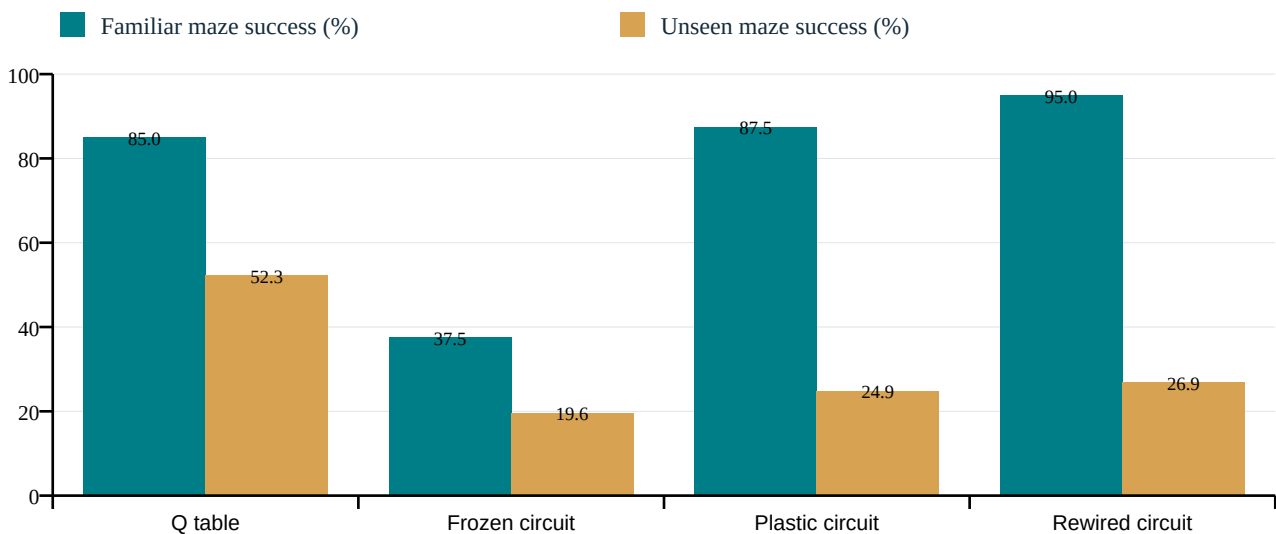
Written after the two full runs completed on 21 September 2026. This section reports observations; the preceding v0.2 amendment was authored before the new benchmark was implemented and run. No test-driven parameter tuning was performed.

Part A: the existing maze controller

Across ten seeds, both reward assignments and 400 online training episodes, fly-probe Q and direct-image Q produced exactly equal Q tables after every episode. Both achieved 2,000/2,000 familiar successes (100%) and 570/2,000 unseen successes (28.5%). Their difference was exactly zero. The neural probe supplies the same decoded tokens and adds no demonstrated value here.

Part B: equal recorded experience

Final success percentages after 2,000 replay updates. Each bar represents 2,000 frozen trials across ten training seeds and both reward assignments. These are point estimates; paired uncertainty is reported on the next page.



Part B uses shared random experience and a wall-aware state representation. Part A uses online training and the original state representation. Compare agents within each part; the different Q-table scores are not a contradiction.

What did training establish?

Connection strengths really changed

The selected anatomical circuit contains 256 Kenyon cells, 97 MBONs and 3,857 recorded positive KC-to-MBON connections. Training changed 3,013 to 3,283 permitted internal strengths per seed/reward condition, as well as the external action readout. Missing edges remained zero and all strengths remained nonnegative. The frozen-circuit control retained its internal weights exactly.

Familiar maze: no clear winner over Q-learning

The plastic circuit scored 87.5% versus 85.0% for the replay Q table: +2.5 percentage points, with a paired seed-bootstrap 95% interval of -12.5 to +17.5 points. That does not establish an advantage. It improved over the frozen-circuit readout at 37.5%, showing that training internal modeled strengths can matter in this particular reduced architecture.

Unseen layouts: the Q table transferred better

The replay Q table scored 52.3%; the plastic circuit scored 24.9%. Plastic minus Q was -27.4 percentage points, with a paired 95% interval of -35.7 to -18.05 points. Random movement scored 29.05%. The trained anatomical model did not demonstrate useful layout transfer in this experiment.

No demonstrated benefit from recorded wiring

The degree-matched rewired network scored 95.0% familiar and 26.9% unseen. Plastic minus rewired was -7.5 points familiar (95% interval -15.0 to 0.0) and -2.0 points unseen (-7.5 to +3.6). The recorded wiring did not outperform this matched control.

Conclusion: current Exp 2 is a tie between two equivalent Q learners. The new reduced plastic circuit learns, but these runs do not show that fly wiring is better than conventional learning. They also do not show that every possible fly-derived model must fail.

Limits, evidence and reproduction

The new model is a reduced feedforward rate network, not the full MaleCNS LIF simulation. Its input projection, tanh dynamics, action readout and gradient/Adam plasticity are engineered. There is no validated dopamine model, natural motor mapping or biological learning claim.

Ten optimization/experience seeds were tested, but only one sampled anatomical subcircuit and one degree-preserving rewiring. Intervals resample seed means after averaging rewarded A/B within each seed; they are not confidence intervals over all possible circuits. Hyperparameters and budgets were fixed rather than optimized separately for every model.

The existing 120-action timeout is treated as terminal, and remaining time is not an observation. This finite-horizon limitation applies to all compared agents. Test layouts are unseen, but absolute position and adjacent walls are observable. No full-map or shortest-path oracle enters a policy.

Approximate mean update-call times per condition on the GX10 host CPU: Q table 0.21 s, frozen-circuit readout 0.59 s, plastic circuit 0.70 s, rewired circuit 0.71 s. These exclude data collection, feature initialization and evaluation. Original full-graph probes used CUDA. Evaluation and immutable checkpoint checks passed.

Reproduce with the README comparison command. The compact, versioned report is docs/research/maze_comparison_results.md; its JSON supplies all learning-curve points and paired intervals. Raw experience, trials, circuit arrays and checkpoints remain outside Git under runs/gx10-a/exp_2_comparison and runs/gx10-b/exp_2_comparison.

Accepted run IDs: 20260921T060909Z-comparison-b15807 (original-controller audit) and 20260921T060903Z-comparison-a72cae (reduced-circuit comparison). Exact scientific source identity is recorded in each manifest and the compact report; immutable releases and a source archive preserve it.

Bookkeeping correction: the first runner hashed its mutable progress pointer before its final update. Explicit finalization_repair.json records retain the original manifests. All 378 immutable artifacts verified after excluding that pointer; no scientific output was changed.